

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-YD-2301-06 Sheet 1 of 1
OPL Title	Outlet Moisture Control & Retention Time (MT-2306)	Discipline	Process / Operations	
Equipment	YD-2301 - POLYMER FLUID BED DRYER	Area / Unit	2300 - POWDER DRYING SYSTEM	
Related Interlock	SEQ-5500 (SIL 2)	P&ID; Ref	TJC-LLD-PID-2301	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **outlet moisture control & retention time (mt-2306)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. The product specification is below 300 ppm moisture. Rising outlet moisture means lost drying and off-spec powder. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the MT-2306 outlet moisture every shift	Permit valid & briefed
2	If moisture rises above 300 ppm, check steam pressure PT-2304 first	Parameter within limit
3	Verify drum speed and hence retention time is correct	Interlock status confirmed
4	Cross-check against the daily lab moisture sample	Change logged on DCS
5	Adjust feed rate or steam to bring moisture back in spec	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- The product specification is below 300 ppm moisture. Rising outlet moisture means lost drying and off-spec powder.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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